

Topic 1.5: Overfitting and Bias–Variance*Lecturer: Dr. Kai Liu**Scribe: ChatGPT*

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Learning goals. Derive expected training and test errors for least squares; explain why reusing training responses gives an optimistic error; prove the bias–variance decomposition; and understand exactly what an expectation over the training dataset means.

1.1 Setup, notation, and sources of randomness

All vectors are columns. For training observations $D = \{(x_i, y_i)\}_{i=1}^n$, let

$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{pmatrix} \in \mathbb{R}^{n \times d}, \quad y = (y_1, \dots, y_n)^\top, \quad y = X\theta^* + \varepsilon.$$

Here $\theta^* \in \mathbb{R}^d$ is fixed and unknown, and the regression function is $f(x) = x^\top \theta^*$. A response includes noise: $y_i = f(x_i) + \varepsilon_i$.

Symbol	Dimension	Meaning
X, y, ε	$n \times d, n, n$	Training inputs, responses, noise
Z, y^{test}, η	$m \times d, m, m$	Test inputs, responses, noise
$\hat{\theta}, \theta^*$	d, d	Fitted and true coefficients
$H = X(X^\top X)^{-1}X^\top$	$n \times n$	Projection (hat) matrix
$x_0, g^D(x_0)$	$d, 1$	One fixed input and its fitted prediction

Assumptions for the least-squares calculations. X has full column rank, so $d \leq n$ and $X^\top X$ is invertible. The conditional noise moments are

$$\mathbb{E}[\varepsilon \mid X] = 0, \quad \text{Cov}(\varepsilon \mid X) = \sigma^2 I_n. \quad (1.1)$$

For a fixed test design Z , let $y^{\text{test}} = Z\theta^* + \eta$, where η has mean zero and covariance $\sigma^2 I_m$ and is independent of the training noise, conditional on the designs. If Z is random, the same moment assumptions are imposed conditional on (X, Z) ; independent sampling of test inputs is sufficient to preserve the training-noise assumptions.

What is being averaged? In Sections 1.2–1.5, we hold the design matrices fixed and average over noise. Later, $\mathbb{E}_D$ averages over the training dataset, including X when the inputs are random. Gaussian noise is sufficient, but the displayed mean and covariance conditions are enough for these calculations. All errors below are *mean squared errors* (MSEs), not square roots. If the model includes an intercept, its column counts toward d .

1.2 Least squares, the hat matrix, and a trace identity

Ordinary least squares (OLS) minimizes

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{2n} \|X\theta - y\|_2^2.$$

Setting the gradient $X^\top(X\theta - y)/n$ to zero gives

$$\hat{\theta} = (X^\top X)^{-1} X^\top y = \theta^* + \underbrace{(X^\top X)^{-1} X^\top}_{A \in \mathbb{R}^{d \times n}} \varepsilon. \quad (1.2)$$

The identity $AX = I_d$ explains why the signal term becomes θ^* . Training predictions are

$$\hat{y} = X\hat{\theta} = Hy, \quad H := X(X^\top X)^{-1} X^\top.$$

1.2.1 Properties of the hat matrix

The matrix H is the orthogonal projector onto the column space of X :

$$\begin{aligned} H^\top &= H, \\ H^2 &= X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1} X^\top = H, \\ HX &= X. \end{aligned}$$

Thus $(I_n - H)X = 0$, and $I_n - H$ is also symmetric and idempotent. Using cyclic invariance of the trace,

$$\text{tr}(H) = \text{tr}((X^\top X)^{-1} X^\top X) = \text{tr}(I_d) = d. \quad (1.3)$$

The two products inside the traces have different sizes; their traces agree.

Lemma 1.1 (Expected quadratic form) *If a random vector u has mean zero and covariance $\sigma^2 I_r$, then for any fixed $r \times r$ matrix B ,*

$$\mathbb{E}[u^\top Bu] = \sigma^2 \text{tr}(B). \quad (1.4)$$

Consequently, for a fixed matrix M with r columns, $\mathbb{E}[\|Mu\|_2^2] = \sigma^2 \text{tr}(MM^\top)$.

Proof. The scalar $u^\top Bu$ equals $\text{tr}(Buu^\top)$. Linearity of expectation gives

$$\mathbb{E}[u^\top Bu] = \text{tr}(B \mathbb{E}[uu^\top]) = \text{tr}(B\sigma^2 I_r) = \sigma^2 \text{tr}(B).$$

For the second statement use $B = M^\top M$ and $\text{tr}(M^\top M) = \text{tr}(MM^\top)$. ■

Condition first when the matrix is random. The lemma applies to matrices such as H after conditioning on X . For example, $\mathbb{E}[\varepsilon^\top H \varepsilon \mid X] = \sigma^2 \text{tr}(H)$. Mean zero alone does not imply $\mathbb{E}[\varepsilon \varepsilon^\top] = \sigma^2 I_n$; the covariance assumption is also needed.

1.3 Expected training error

The realized training MSE is

$$\widehat{R}_{\text{train}} := \frac{1}{n} \|X\hat{\theta} - y\|_2^2. \quad (1.5)$$

This quantity depends on the particular noise vector in the observed dataset.

1.3.1 Step 1: The residual contains only part of the noise

Since $HX = X$,

$$\begin{aligned} X\hat{\theta} - y &= Hy - y \\ &= (H - I_n)(X\theta^* + \varepsilon) \\ &= (H - I_n)\varepsilon. \end{aligned}$$

Therefore

$$\widehat{R}_{\text{train}} = \frac{1}{n} \varepsilon^\top (I_n - H)^\top (I_n - H) \varepsilon = \frac{1}{n} \varepsilon^\top (I_n - H) \varepsilon.$$

The last equality uses symmetry and idempotence of $I_n - H$.

1.3.2 Step 2: Average over training noise

Applying Lemma 1.1 conditional on X ,

$$\begin{aligned} \mathbb{E}[\widehat{R}_{\text{train}} \mid X] &= \frac{\sigma^2}{n} \text{tr}(I_n - H) \\ &= \frac{\sigma^2}{n} (n - d) = \boxed{\sigma^2 \left(1 - \frac{d}{n}\right)}. \end{aligned} \quad (1.6)$$

If X is random and has full column rank almost surely, averaging this constant over X gives the same unconditional expectation.

1.3.3 Why is the result smaller than the noise variance?

The fitted responses are

$$\hat{y} = X\theta^* + H\varepsilon.$$

OLS fits the true signal *and* the component $H\varepsilon$ of training noise. Only $(I_n - H)\varepsilon$ remains in the residual. The orthogonal decomposition

$$\varepsilon = H\varepsilon + (I_n - H)\varepsilon$$

has expected squared lengths $\sigma^2 d$ and $\sigma^2(n - d)$, respectively. Thus using the same responses to fit and evaluate the model hides some noise.

Numerical example. If $n = 100$, $d = 20$, and $\sigma^2 = 1$, then $\mathbb{E}[\widehat{R}_{\text{train}} \mid X] = 0.8$. This is an expectation; one observed training MSE need not equal 0.8.

Boundary and estimation. When $n = d$ and X is invertible, $H = I_n$ and the training MSE is exactly zero. For $n > d$, the quantity $\|y - X\hat{\theta}\|_2^2 / (n - d)$ is an unbiased estimator of σ^2 under the stated model. Dividing by n instead gives training MSE.

1.4 Fresh responses at the same inputs

To isolate the effect of reusing training responses, take the test design to be $Z = X$, with $m = n$, but generate *new responses*

$$y^{\text{new}} = X\theta^* + \eta.$$

Conditional on X , η is independent of ε , has mean zero, and has covariance $\sigma^2 I_n$. The fitted model still uses only $y = X\theta^* + \varepsilon$.

Define the realized MSE against these fresh responses:

$$\widehat{R}_{\text{new}} := \frac{1}{n} \left\| X\hat{\theta} - y^{\text{new}} \right\|_2^2 = \frac{1}{n} \|H\varepsilon - \eta\|_2^2.$$

Here the old fitted noise $H\varepsilon$ cannot cancel the new noise η .

1.4.1 Expand and average each term

$$n\widehat{R}_{\text{new}} = \varepsilon^\top H^\top H \varepsilon + \eta^\top \eta - 2\eta^\top H \varepsilon.$$

The three conditional expectations are

$$\mathbb{E}[\varepsilon^\top H^\top H \varepsilon \mid X] = \sigma^2 \text{tr}(H) = \sigma^2 d, \quad \mathbb{E}[\eta^\top \eta \mid X] = n\sigma^2,$$

and

$$\mathbb{E}[\eta^\top H \varepsilon \mid X] = \mathbb{E}_{\varepsilon \mid X} \left[\underbrace{\mathbb{E}[\eta^\top \mid X, \varepsilon]}_0 H \varepsilon \right] = 0.$$

Thus

$$\mathbb{E}[\widehat{R}_{\text{new}} \mid X] = \sigma^2 \left(1 + \frac{d}{n} \right). \quad (1.7)$$

The corresponding expected gap is

$$\mathbb{E}[\widehat{R}_{\text{new}} - \widehat{R}_{\text{train}} \mid X] = \frac{2\sigma^2 d}{n}. \quad (1.8)$$

Continuing the example. With $n = 100$, $d = 20$, and $\sigma^2 = 1$, expected training MSE is 0.8, whereas expected fresh-response MSE at the same inputs is 1.2. The expected gap is 0.4.

Same inputs do not mean the same test observations. This calculation reuses X but not the training responses. If we also reused the training noise, we would be measuring training error. The formula $\sigma^2(1 + d/n)$ is specific to this same-design experiment; new input locations require the calculation on the next page.

1.5 Expected test error at general input locations

Let $Z \in \mathbb{R}^{m \times d}$ contain arbitrary fixed test inputs and let $y^{\text{test}} = Z\theta^* + \eta$. The test MSE is

$$\hat{R}_{\text{test}} = \frac{1}{m} \left\| Z\hat{\theta} - y^{\text{test}} \right\|_2^2.$$

Keep the normalizing factor $1/m$: the number of test observations need not equal the number n of training observations.

1.5.1 Step 1: Express the prediction error in terms of noise

From (1.2),

$$Z\hat{\theta} - y^{\text{test}} = \underbrace{Z(X^\top X)^{-1}X^\top}_{B \in \mathbb{R}^{m \times n}} \varepsilon - \eta.$$

Conditional on (X, Z) , B is fixed and the two noises are independent. Thus

$$\begin{aligned} \mathbb{E}[\hat{R}_{\text{test}} \mid X, Z] &= \frac{1}{m} \mathbb{E}[\|B\varepsilon\|_2^2 + \|\eta\|_2^2 - 2\eta^\top B\varepsilon \mid X, Z] \\ &= \frac{\sigma^2}{m} \text{tr}(BB^\top) + \sigma^2. \end{aligned}$$

The cross term vanishes by the same conditional-expectation argument as in Section 1.4.

1.5.2 Step 2: Simplify the matrix product

$$BB^\top = Z(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}Z^\top = Z(X^\top X)^{-1}Z^\top.$$

Consequently,

$$\boxed{\mathbb{E}[\hat{R}_{\text{test}} \mid X, Z] = \underbrace{\sigma^2}_{\text{fresh response noise}} + \underbrace{\frac{\sigma^2}{m} \text{tr}(Z(X^\top X)^{-1}Z^\top)}_{\text{uncertainty in the fitted predictor}}}. \quad (1.9)$$

Equivalently, this equals $\frac{\sigma^2}{m} \text{tr}(Z(X^\top X)^{-1}Z^\top + I_m)$.

1.5.3 Checks and interpretation

If $Z = X$ and $m = n$, the trace is $\text{tr}(H) = d$, recovering (1.7). For general Z , it need not equal d . Writing the rows of Z as $z_j^\top$ gives

$$\text{tr}(Z(X^\top X)^{-1}Z^\top) = \sum_{j=1}^m z_j^\top (X^\top X)^{-1} z_j.$$

Prediction uncertainty is larger in directions that the training inputs estimate poorly. Against the *noiseless* targets $Z\theta^*$, the expected MSE is only the second term in (1.9); the extra σ^2 comes from testing against noisy responses.

For random test inputs and random X , take an outer expectation of the right side, provided it is finite. Do not replace $(X^\top X)^{-1}$ by $(\mathbb{E}[X^\top X])^{-1}$; inverse and expectation do not generally commute.

1.6 The bias–variance decomposition at a fixed input

Now allow any regression function f and any fitted predictor g^D . Fix a test input x_0 and write its fresh response as

$$Y_0 = f(x_0) + \xi, \quad \mathbb{E}[\xi \mid D, x_0] = 0, \quad \mathbb{E}[\xi^2 \mid D, x_0] = \sigma^2.$$

Independent zero-mean test noise with variance σ^2 satisfies these conditions. Assume the prediction has a finite second moment. The scalar ξ is fresh test noise; it is distinct from the training noise vector ε .

Define the average predictor and its bias and variance:

$$\begin{aligned} \bar{g}(x_0) &:= \mathbb{E}_D[g^D(x_0)], \\ \text{Bias}(x_0) &:= \bar{g}(x_0) - f(x_0), \\ \text{Var}_D(g^D(x_0)) &:= \mathbb{E}_D[(g^D(x_0) - \bar{g}(x_0))^2]. \end{aligned}$$

Here $\bar{g}(x_0)$ is an average over hypothetical training datasets, not an average over the input locations in one dataset.

Theorem 1.2 (Bias–variance decomposition) *Under these assumptions, the expected squared prediction error is*

$$\boxed{R(x_0) := \mathbb{E}_{D,\xi}[(g^D(x_0) - Y_0)^2] = \sigma^2 + \text{Var}_D(g^D(x_0)) + \text{Bias}(x_0)^2.} \quad (1.10)$$

Proof. Set $a = g^D(x_0) - \bar{g}(x_0)$ and $b = \bar{g}(x_0) - f(x_0)$. Then a is random, b is fixed, $\mathbb{E}_D[a] = 0$, and

$$g^D(x_0) - Y_0 = a + b - \xi.$$

Expanding the square gives

$$(a + b - \xi)^2 = a^2 + b^2 + \xi^2 + 2ab - 2a\xi - 2b\xi.$$

Each cross term has zero expectation:

$$\mathbb{E}[ab] = b \mathbb{E}[a] = 0, \quad \mathbb{E}[b\xi] = b \mathbb{E}[\xi] = 0, \quad \mathbb{E}[a\xi] = \mathbb{E}_D[a \underbrace{\mathbb{E}[\xi \mid D, x_0]}_0] = 0.$$

The remaining terms are variance, squared bias, and noise variance. ■

Two zero means do not guarantee a zero product expectation. If $a = \xi$ and $\text{Var}(\xi) > 0$, then $\mathbb{E}[a] = \mathbb{E}[\xi] = 0$ but $\mathbb{E}[a\xi] = \mathbb{E}[\xi^2] > 0$. The cross term above vanishes because the *fresh* test noise has zero conditional mean given the training data.

The final term is **squared bias**, not bias. If the noise variance depends on the input, replace σ^2 by $\sigma^2(x_0)$. For overall prediction risk, average the pointwise decomposition over an independent test input X_0 .

1.7 What does expectation over the training dataset mean?

For random training inputs, the dataset is equivalently $D = (X, y)$. We average using the conditional response model $y = X\theta^* + \varepsilon$. For any integrable function of the dataset, the law of total expectation says

$$\boxed{\mathbb{E}_D[\text{quantity}] = \mathbb{E}_X[\mathbb{E}_{y|X}[\text{quantity}]]} . \quad (1.11)$$

The inner expectation holds X fixed and varies the responses. The outer expectation then varies the training inputs.

1.7.1 Apply the two-stage calculation to least squares

For a fixed test input x_0 , the OLS prediction is

$$g^D(x_0) = x_0^\top \hat{\theta} = x_0^\top (X^\top X)^{-1} X^\top y.$$

Stage 1: Hold the training inputs fixed. The conditional mean of the responses is

$$\mathbb{E}[y | X] = X\theta^* + \underbrace{\mathbb{E}[\varepsilon | X]}_0 = X\theta^*.$$

Since the matrix multiplying y is fixed conditional on X ,

$$\begin{aligned} \mathbb{E}_{y|X}[g^D(x_0)] &= x_0^\top (X^\top X)^{-1} X^\top \mathbb{E}[y | X] \\ &= x_0^\top \underbrace{(X^\top X)^{-1} X^\top X}_{I_d} \theta^* = x_0^\top \theta^*. \end{aligned} \quad (1.12)$$

Stage 2: Average over the training inputs. Apply (1.11):

$$\bar{g}(x_0) = \mathbb{E}_D[g^D(x_0)] = \mathbb{E}_X[x_0^\top \theta^*] = \boxed{x_0^\top \theta^* = f(x_0)}. \quad (1.13)$$

The final equality holds because x_0 and θ^* are fixed: $x_0^\top \theta^*$ does not depend on the random training matrix X . For random designs, assume full column rank almost surely and the moments needed for these expectations.

1.7.2 Interpretation and a common pitfall

Imagine collecting many training datasets, fitting OLS separately to each, and recording every fitted prediction at the *same* input x_0 . Their expected value is $f(x_0)$. This is the meaning of unbiased prediction; an individual fitted model can still be far from $f(x_0)$. If X is fixed by the experiment, only Stage 1 is needed.

With random X , $\mathbb{E}[\varepsilon] = 0$ by itself is insufficient. The random matrix $(X^\top X)^{-1} X^\top$ can be correlated with the noise. The condition $\mathbb{E}[\varepsilon | X] = 0$ addresses this issue. Independence of ε and X , together with mean-zero noise, is sufficient, but independence is stronger than necessary.

What is recovered on average? OLS recovers the noiseless regression function $f(x_0)$ in expectation. The future response is $Y_0 = f(x_0) + \xi$; its particular fresh noise realization cannot be recovered from the training data.

1.8 Variance of OLS and a fully worked example

Unbiasedness removes the squared-bias term; it does not remove variance. Let $G = X^\top X$ and $A = G^{-1}X^\top$. Conditional on X ,

$$\begin{aligned}\text{Cov}(\hat{\theta} \mid X) &= A \text{Cov}(\varepsilon \mid X) A^\top \\ &= \sigma^2 G^{-1} (X^\top X) G^{-1} = \sigma^2 G^{-1}.\end{aligned}\tag{1.14}$$

Consequently, for a fixed input x_0 ,

$$\boxed{\text{Var}(g^D(x_0) \mid X) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0.}\tag{1.15}$$

Because the conditional bias is zero, the expected error against a fresh response at x_0 is

$$\mathbb{E}[(g^D(x_0) - Y_0)^2 \mid X] = \sigma^2 + \sigma^2 x_0^\top (X^\top X)^{-1} x_0.$$

This is the one-point version of (1.9). At training input x_i , the quadratic form is H_{ii} , and $\sum_i H_{ii} = \text{tr}(H) = d$. Averaging the prediction variances at those inputs therefore gives $\sigma^2 d/n$.

1.8.1 If the training inputs are random

The law of total variance gives

$$\begin{aligned}\text{Var}_D(g^D(x_0)) &= \mathbb{E}_X[\text{Var}(g^D(x_0) \mid X)] + \text{Var}_X(\mathbb{E}[g^D(x_0) \mid X]) \\ &= \sigma^2 \mathbb{E}_X[x_0^\top (X^\top X)^{-1} x_0] + 0.\end{aligned}$$

The second term is zero because the conditional mean is the fixed number $x_0^\top \theta^*$. This identity assumes a finite second moment; almost-sure invertibility alone does not ensure a finite expected inverse.

1.8.2 Example: estimating one constant mean

Suppose $y_i = \theta^* + \varepsilon_i$, with independent noise of mean zero and variance σ^2 . Here $X = \mathbf{1}_n$, $d = 1$, and

$$\hat{\theta} = \bar{y} = \theta^* + \frac{1}{n} \sum_{i=1}^n \varepsilon_i, \quad \mathbb{E}[\hat{\theta}] = \theta^*, \quad \text{Var}(\hat{\theta}) = \frac{\sigma^2}{n}.$$

For a new observation $Y_0 = \theta^* + \xi$ independent of the training data,

$$\mathbb{E}[(\bar{y} - Y_0)^2] = \underbrace{\sigma^2}_{\text{new noise}} + \underbrace{\sigma^2/n}_{\text{estimation variance}} + \underbrace{0}_{\text{squared bias}}.$$

For $n = 10$ and $\sigma^2 = 1$, expected training MSE is 0.9 and expected test MSE is 1.1, although the estimator is unbiased. More training samples reduce estimation variance; the fresh-response noise variance remains 1.

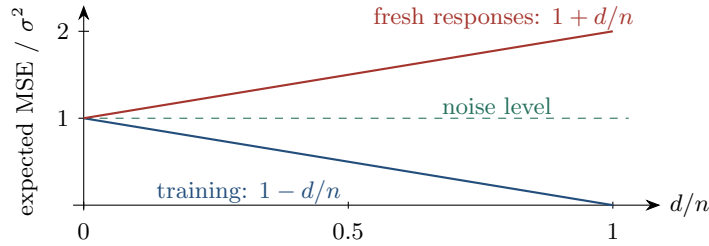
1.9 What these calculations say about overfitting

Overfitting occurs when fitting the training observations also captures sample-specific noise, so improved training fit does not translate into improved prediction on fresh observations. In our correctly specified OLS calculation, the fitted noise is $H\varepsilon$.

1.9.1 The exact same-design comparison

For full-column-rank models that contain the true regression function,

$$\frac{\mathbb{E}[\widehat{R}_{\text{train}} | X]}{\sigma^2} = 1 - \frac{d}{n}, \quad \frac{\mathbb{E}[\widehat{R}_{\text{new}} | X]}{\sigma^2} = 1 + \frac{d}{n}.$$



The lines illustrate the formulas as a function of the ratio d/n ; actual model dimensions are integers. They concern fresh responses at the training inputs, rather than arbitrary new input locations. They show that an unbiased method can have an increasing test error because its prediction variance increases.

1.9.2 Where does approximation bias enter?

The earlier zero-bias result assumes $f(x) = x^\top \theta^*$ in the chosen features. A model that omits relevant structure can have nonzero bias. For fixed inputs, set $f_X = (f(x_1), \dots, f(x_n))^\top$, which need not lie in the column space of X . Then $\mathbb{E}[\hat{y} | X] = Hf_X$ and define

$$B_X^2 := \frac{1}{n} \|(I_n - H)f_X\|_2^2.$$

The same residual expansions, now retaining the signal residual, give

$$\begin{aligned} \mathbb{E}[\widehat{R}_{\text{train}} | X] &= B_X^2 + \sigma^2(1 - d/n), \\ \mathbb{E}[\widehat{R}_{\text{new}} | X] &= B_X^2 + \sigma^2(1 + d/n). \end{aligned}$$

For nested feature spaces at the same inputs, B_X^2 cannot increase as the space expands, while d/n increases. These opposing effects explain a possible bias–variance trade-off; test error need not have a U-shape in every learning problem.

Scope. These inverse-based OLS formulas assume $d \leq n$ and full column rank. They do not cover over-parameterized minimum-norm fitting when $d > n$. An expected training–test gap also does not imply the same ordering for every realized pair of datasets.

1.10 Optional extension: ridge makes the trade-off explicit

Fix X and consider a penalty on *all* coordinates:

$$\hat{\theta}_\lambda = \arg \min_{\theta} \left\{ \frac{1}{2} \|X\theta - y\|_2^2 + \frac{\lambda}{2} \|\theta\|_2^2 \right\} = (G + \lambda I_d)^{-1} X^\top y, \quad G = X^\top X.$$

The data-fit term is not divided by n here; this specifies the scale of λ . Unlike the unregularized formulas, $\lambda > 0$ makes this inverse well-defined even when X lacks full column rank.

1.10.1 Conditional bias and variance at a fixed test input

Using $G = (G + \lambda I_d) - \lambda I_d$,

$$\mathbb{E}[\hat{\theta}_\lambda \mid X] = (G + \lambda I_d)^{-1} G \theta^* = \theta^* - \lambda (G + \lambda I_d)^{-1} \theta^*.$$

Thus the prediction bias is

$$b_\lambda(x_0; X) := \mathbb{E}[x_0^\top \hat{\theta}_\lambda \mid X] - x_0^\top \theta^* = -\lambda x_0^\top (G + \lambda I_d)^{-1} \theta^*.$$

Propagating the noise covariance gives the prediction variance

$$v_\lambda(x_0; X) = \sigma^2 x_0^\top (G + \lambda I_d)^{-1} G (G + \lambda I_d)^{-1} x_0.$$

Against a fresh noisy response, conditional risk is

$$\boxed{R_\lambda(x_0 \mid X) = \sigma^2 + v_\lambda(x_0; X) + b_\lambda(x_0; X)^2.} \quad (1.16)$$

Regularization can reduce variance at the cost of introducing bias. It does not guarantee a lower total risk for every λ or signal.

1.10.2 A numerical example with a penalized constant mean

Return to $X = \mathbf{1}_n$. This toy example deliberately penalizes the constant coefficient; some software leaves an intercept unpenalized. Then

$$\hat{\theta}_\lambda = \frac{n}{n + \lambda} \bar{y}, \quad b_\lambda = -\frac{\lambda}{n + \lambda} \theta^*, \quad v_\lambda = \frac{\sigma^2 n}{(n + \lambda)^2}.$$

For $n = 10$, $\sigma^2 = 1$, and $\theta^* = 0.2$, the expected error for a new response is $1 + b_\lambda^2 + v_\lambda$:

λ	Squared bias	Variance	Expected test MSE
0	0.00000	0.10000	1.10000
10	0.01000	0.02500	1.03500
25	0.02041	0.00816	1.02857
$\lambda \rightarrow \infty$	0.04000	0.00000	1.04000

Here some shrinkage improves prediction even though OLS is unbiased. The numbers use known parameters for illustration; in practice, select regularization using validation data or cross-validation, keeping the final test data separate from model selection.

1.11 Summary and checks for understanding

1.11.1 Which expectation and which target?

The distinction between an observed loss and an expected loss matters: $\hat{R}_{\text{train}}$ and $\hat{R}_{\text{test}}$ are random quantities; the formulas below describe their expectations.

Setting	Result
OLS, training responses	$\mathbb{E}[\hat{R}_{\text{train}} \mid X] = \sigma^2(1 - d/n)$
OLS, fresh responses at the same inputs	$\mathbb{E}[\hat{R}_{\text{new}} \mid X] = \sigma^2(1 + d/n)$
OLS, general test design Z	$\mathbb{E}[\hat{R}_{\text{test}} \mid X, Z] = \sigma^2 + \frac{\sigma^2}{m} \text{tr}(Z(X^\top X)^{-1}Z^\top)$
Any predictor, fixed test input	$R(x_0) = \sigma^2 + \text{Var}_D(g^D(x_0)) + \text{Bias}(x_0)^2$
Correctly specified OLS, averaging over training datasets	$\bar{g}(x_0) = \mathbb{E}_X[\mathbb{E}_{y X}[g^D(x_0)]] = x_0^\top \theta^*$

The OLS rows use the full-rank, correct-model, and noise assumptions from Section 1.1; the bias–variance row uses the fresh-noise and moment assumptions in Section 1.6.

1.11.2 Questions for discussion

1. Why are the rows of X written as $x_i^\top$? Check the dimensions of $\hat{\theta}$, H , and $Z(X^\top X)^{-1}Z^\top$.
2. Show that $H^\top = H$, $H^2 = H$, and $\text{tr}(H) = d$. Which step requires full column rank?
3. Explain why $\mathbb{E}[\varepsilon] = 0$ is not enough to evaluate $\mathbb{E}[\varepsilon^\top B \varepsilon]$.
4. In the same-design test experiment, what is reused and what must be generated independently? Why does the cross term vanish?
5. If $n = 80$, $d = 8$, and $\sigma^2 = 2$, compute expected training MSE, same-design fresh-response MSE, and their expected gap.
6. Derive the bias–variance decomposition by expanding $(a + b - \xi)^2$. Explain separately why each cross term disappears.
7. Reproduce the two-stage expectation in Section 1.7. Why may the matrix be taken outside the inner expectation?
8. Does zero bias imply zero test error? Explain using the constant-mean example and identify which term decreases with n .
9. Why is $\sigma^2(1 + d/n)$ not a general formula for arbitrary test inputs? Which matrix replaces H in that calculation?
10. Can a biased estimator predict better than an unbiased one? Use the ridge example to explain your answer.